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Editors

*Machine Learning and Knowledge Extraction* (MAKE)

MDPI

Dear Editors,

We are pleased to submit our manuscript entitled

**“Intrinsic-Dimension-Guided Bottleneck Selection for Autoencoders and Variational Autoencoders with Active-Unit Diagnostics”**

by Chika Obata, Mizuki Dai, and Kenya Jin’no, for consideration as a Research Article in the *Learning* section of *Machine Learning and Knowledge Extraction* (MAKE).

**Significance and context.** The bottleneck dimension of autoencoders (AEs) and variational autoencoders (VAEs) is one of their most fundamental design parameters, yet in practice it is almost always set by convention or trial and error. Existing work has measured the intrinsic dimension of data and of deep representations (e.g., Facco et al. 2017; Pope et al. 2021; Ansuini et al. 2019) or analyzed posterior collapse in VAEs (Lucas et al. 2019), but stops short of connecting these threads into a design procedure. Our manuscript closes this gap with a conditional, diagnostic estimate–set–verify protocol: an intrinsic-dimension estimate of the input data serves as a search-window reference for the bottleneck dimension, and post-training verification based on active units (AU) and reconstruction error diagnoses whether the chosen dimension behaved as intended. The protocol is motivated theoretically by a topological lower bound and a linear-Gaussian rate–distortion surrogate, validated with multiple seeds on synthetic manifolds with known ground truth and on MNIST and Fashion-MNIST, and reduces the training runs of a full grid search by roughly 60–80% on the tested MNIST grid while containing the best observed linear-probe accuracy. Importantly, the paper also quantifies where the approach reaches its limits (natural images and convolutional VAEs), rather than claiming a universal solution.

**Fit to the scope of MAKE.** The paper extracts actionable knowledge from learned representations—reading effective latent dimensionality from trained models—and turns it into a transparent, reproducible design procedure with pre-specified decision rules and a failure-mode taxonomy. This combination of machine learning methodology and knowledge extraction, together with the emphasis on reproducibility (all code, configurations, and seed-specific logs are provided as an anonymized supplementary archive for review, to be released publicly with a DOI upon acceptance), aligns directly with the aims of MAKE.

A preliminary analysis of part of this study was presented at NOLTA 2025 (Obata & Jin’no, “Analysis of the Relationship Between Manifold Structure and Latent Space Capacity of Input Data Using VAE”); it is cited in the manuscript, which substantially extends it with the full protocol, multi-seed validation, and applicability analysis. The manuscript has not previously been submitted to MAKE or to any other MDPI journal. The use of Claude Code (Anthropic; Claude Fable 5 model) for assistance in generating experimental program code, English grammar checking, and proofreading is disclosed with reasons in the Acknowledgments section, and the authors take full responsibility for the content. The authors declare no conflicts of interest.

We confirm that neither the manuscript nor any parts of its content are currently under consideration for publication with or published in another journal.

All authors have approved the manuscript and agree with its submission to MAKE.

Thank you for considering our manuscript. We look forward to the reviewers’ feedback.

Sincerely,

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